Machine Learning Practical Assignment

Titanic Survival Dataset - Preprocessing and Classification

Algorithms: K-Nearest Neighbors (KNN), Decision Tree, and Random Forest

Dataset

- Use the Titanic - Machine Learning from Disaster dataset and work with train.csv because it contains the target column Survived (0 = did not survive, 1 = survived).
  Kaggle dataset page: https://www.kaggle.com/competitions/titanic/data

Part A - Data Understanding and Preprocessing

- Load train.csv and inspect the dataset using head(), shape, info(), and basic statistics.
- Check missing values and identify numerical and categorical features.
- Remove columns that are not useful for the model, and handle missing values appropriately.
- Encode the categorical variables required for machine learning.
- Create the final feature matrix X and target vector y.

Part B - Train/Test Split and Scaling

- Split the data into 80% training and 20% testing using random_state = 42 and stratify = y.
- For KNN, standardize the required numerical features using StandardScaler. Fit the scaler on the training data only.
- Briefly state why scaling is important for KNN but usually not required for Decision Tree and Random Forest.

Part C - K-Nearest Neighbors (KNN)

- Train a KNN classifier starting with k = 5.
- Try a small set of k values (for example: 3, 5, 7, and 9) and select the best one.
- Evaluate the final KNN model using Accuracy, Precision, Recall, F1-score, and the Confusion Matrix.

Part D - Decision Tree

- Train a DecisionTreeClassifier using the same train/test split.
- Compare the unrestricted tree with at least two max_depth values and observe the effect on train and test accuracy.
- Choose a final tree and report Accuracy, Precision, Recall, F1-score, and the Confusion Matrix.

Part E - Random Forest

- Train a RandomForestClassifier with at least 100 trees.
- Evaluate it using Accuracy, Precision, Recall, F1-score, and the Confusion Matrix.
- Display the five most important features.

Part F - Model Comparison

- Compare the test performance of KNN, Decision Tree, and Random Forest.
- State which model performed best and give a short reason for your choice.

End of Assignment